

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AMA578
Subject Title	Emerging Topics in Mathematics and AI
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	N.A.
Objectives	The objective of this course is to introduce students to emerging topics at the intersection of applied mathematics and artificial intelligence (AI). Specifically, the course aims to: 1. Expose students to recent advances and cutting-edge developments in artificial intelligence and applied mathematics, with a focus on emerging areas. 2. Examine contemporary techniques and methodologies in both artificial intelligence and applied mathematics. 3. Equip students with practical skills and tools to develop AI models and perform data analysis across various disciplines.
Intended Learning Outcomes	Upon completion of this course, students will be able to: a) Demonstrate a deep understanding of advanced concepts and challenges at the intersection of applied mathematics, artificial intelligence, and data science in emerging areas. b) Apply mathematical modeling and analytical techniques, alongside AI methods, to address complex problems in cutting-edge applications. c) Utilize mathematical and AI tools to analyze various forms of big data, and interpret the results in a visualizable way. d) Design and implement AI-driven solutions to novel real-world challenges, integrating mathematical reasoning and considering diverse requirements and constraints. e) Collaborate effectively in teams, and communicate technical findings through presentations and written reports, with an emphasis on the mathematical foundations of AI solutions.
Subject Synopsis/ Indicative Syllabus	This course explores emerging topics at the intersection of applied mathematics and artificial intelligence (AI), with a focus on their applications across diverse scientific and industrial domains. Students will be introduced to advanced mathematical concepts and state-of-the-art AI techniques and will learn how these tools are transforming fields such as science, mathematics, finance, and healthcare. Through lectures, case studies, and hands-on projects, students will gain both theoretical understanding and practical experience in applying mathematical and AI methods to solve real-world problems.

	Indicative Syllabus: - Overview of applied mathematics in the AI era - Historical context and recent breakthroughs - Ethical and societal considerations - Mathematical Foundations for AI - Linear algebra, optimization, and probability theory - Deep learning architectures and mathematical underpinnings - Explainable AI and interpretability - AI for Science and Engineering - AI-driven scientific discovery and hypothesis generation - Mathematical modeling of physical and biological systems - AI for Mathematics - Automated theorem proving and symbolic computation - Machine learning for mathematical conjecture and discovery - AI-assisted mathematical research and education - AI for Finance - Mathematical models in quantitative finance - AI for risk assessment, algorithmic trading, and fraud detection - Time series analysis and financial forecasting - AI for Healthcare - Mathematical modeling in epidemiology and medical imaging - AI for diagnostics, personalized medicine, and drug discovery - Data privacy and ethical considerations in healthcare AI - Interdisciplinary Case Studies and Applications - Recent research highlights and open problems						
Teaching/Learning Methodology	<u>Lectures:</u> Introduce emerging AI topics, tools, and interdisciplinary applications. <u>Tutorials:</u> Cases studies to demonstrate the practical use of AI models in different application disciplines.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Assignments	20%	✓	✓	✓	✓	✓
	2. Project	30%	✓	✓	✓	✓	✓
	3. Exam	50%	✓	✓	✓	✓	✓
	Total	100%					
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessments consist of assignments and a project, which are designed to facilitate students to achieve the intended learning outcomes. The project is designed to enhance students' ability to acquire the understanding and use different knowledge, principles, techniques, and tools to solve a real problem through a team. Assignments and tests are to ensure the students understand the concepts. The examination will evaluate students' understanding and usage of AI and							

	applied mathematical knowledge.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	13 Hrs.
	Other student study effort:	
	▪ Self-learning from reference materials	35 Hrs.
	▪ Reading and research for the case study	40 Hrs.
	Total student study effort	114 Hrs.
Reading List and References	[1] Goodfellow, I., Bengio, Y., & Courville, A. (2016). <i>Deep Learning</i>. MIT Press. [2] Szepesvári, C. (2022). <i>Algorithms for reinforcement learning</i>. Springer nature. [3] Shaheen, M. Y. (2021). Applications of Artificial Intelligence (AI) in healthcare: A review. <i>Science Open Preprints</i>. [4] Smith, P. D. (2018). <i>Hands-On Artificial Intelligence for Beginners: An introduction to AI concepts, algorithms, and their implementation</i>. Packt Publishing Ltd. [5] Wang, H., Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., ... & Zitnik, M. (2023). Scientific discovery in the age of artificial intelligence. <i>Nature</i>, 620(7972), 47-60.	